

Bragg Scattering of Light in Vacuum Structured by Strong Periodic Fields

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Elastic scattering of laser radiation due to vacuum polarization by spatially modulated strong electromagnetic fields is considered. The Bragg interference arising at a specific impinging direction of the probe wave concentrates the scattered light in specular directions. The interference maxima are enhanced with respect to the usual vacuum polarization effect proportional to the square of the number of modulation periods within the interaction region. The Bragg scattering can be employed to detect the vacuum polarization effect in a setup of multiple crossed superstrong laser beams with parameters envisaged in the future Extreme Light Infrastructure.

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Quantum electrodynamical vacuum is polarizable in strong external fields. The critical field when the vacuum polarization effects become significant is $E_{\text{cr}} = m^2/e$ [1], where e and m are the electron charge and mass, respectively, and $I_{\text{cr}} \equiv E_{\text{cr}}^2/8\pi \approx 2.3 \times 10^{29} \text{ W/cm}^2$, $\hbar = c = 1$ units are used throughout. The only vacuum polarization effect observed experimentally is the Delbrück scattering [2]. In this process vacuum is polarized due to a singular Coulomb field of highly charged ions. In contrast to that, vacuum polarization effects are not observed in macroscopic electromagnetic fields created in a laboratory. Since 2000, the experiments of the PVLAS collaboration that have been underway [3] are devoted to the measurement of vacuum birefringence in a constant magnetic field, the latter reaching $B/B_{\text{cr}} \sim 10^{-9}$, where $B_{\text{cr}} = m^2/e = 4.4 \times 10^9 \text{ T}$ is the critical magnetic field. Recently, strong field laser technique is advancing rapidly, fostered, from one side, by the laser fusion program [4] and, from another side, by the Extreme Light Infrastructure (ELI) project [5]. The laser field at ELI will be the strongest field created in a laboratory (with intensities up to $I \sim 10^{26} \text{ W/cm}^2$) which can be harnessed to test the strong field QED theory via vacuum polarization effects [6,7].

Even in the strongest laser fields the vacuum polarization effects are perturbative because of the smallness of the characteristic parameter $E/E_{\text{cr}} \ll 1$. In terms of Feynman diagrams, the largest contribution to vacuum polarization arises from the box diagram, see Fig. 1(a). The depicted diagram, with two legs belonging to a constant uniform magnetic field or to an external laser field with a uniform amplitude, describes the scattering to zero angle that induces the vacuum refractive index. Thus, in these cases, vacuum behaves as a uniform birefringent medium [8]. Possibilities to observe vacuum birefringence in strong laser fields have been recently discussed in [7,9]. The box diagram with one leg describing the external field does not exist in a constant uniform magnetic field [10],

while in a laser field it describes the photon-photon scattering process [11]. The attempts to observe photon-photon scattering with laser beams were only able to determine the upper limit of the cross section [12]. To enhance the cross section, a third laser beam can be applied to stimulate photon-photon scattering [13]. The possibility of photon-photon scattering with modern strong laser beams is analyzed in [14–16]. In the external field nonuniform in space, vacuum behaves as a nonuniform medium and photon scattering (or diffraction) becomes possible. The vacuum polarization effects due to a spatial gradient of magnetic field are considered in [17]. Diffraction effects due to the vacuum nonuniform polarization in strong focused laser beams are investigated in [18]. In optics of continuous media, it is known that the periodic structure of a medium can significantly enhance the scattering due to interference of the scattered light generated from different layers of the structure (Bragg scattering) [19]. The Bragg concept is quite general and is applied not only in optics [20] but also in quantum optics [21], atom optics [22] and for matter waves [23].

In this Letter, we investigate Bragg scattering of a probe laser beam due to vacuum polarization in a strong spatially modulated external electromagnetic field. At certain

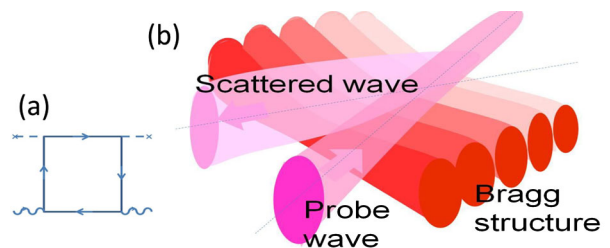


FIG. 1 (color online). (a) The Feynman diagram describing the photon scattering in a strong external electromagnetic field. (b) Bragg scattering of a probe laser beam by a set of focused strong laser beams.

scattering angles when the Bragg interference condition is fulfilled, the probability of the photon scattering is enhanced with respect to the usual photon-photon scattering by a factor proportional to the square of the number of periods in the structure. The enhancement is maintained also in the total probability of the scattering, integrated by the scattering angle. First, we show the effect on the theoretically more transparent case of a spatially periodic magnetic field of a magnetic undulator. Then, we discuss the Bragg scattering when a probe laser beam penetrates the periodic structure of multiple focused laser beams, see Fig. 1(b). We consider the experimental realization of the Bragg scattering in the future ELI facility and its advantage with respect to stimulated photon-photon scattering [15].

We consider photon scattering in an external field, i.e., the transition $|1(\mathbf{k}_1, \mathbf{e}_1)\rangle \rightarrow |1(\mathbf{k}_2, \mathbf{e}_2)\rangle$, where $\mathbf{k}_{1,2}$ and $\mathbf{e}_{1,2}$ are the momentum and polarization of the incoming and outgoing photons, respectively. The transition matrix element reads:

$$T_{21} = -i\langle 1(\mathbf{k}_2, \mathbf{e}_2) | \int_{-\infty}^{\infty} dt d^3\mathbf{r} j_{\mu}(\mathbf{r}, t) A^{\mu}(\mathbf{r}, t) | 1(\mathbf{k}_1, \mathbf{e}_1) \rangle, \quad (1)$$

where $A^{\mu}(\mathbf{r})$ is the field vector potential and $j_{\mu}(\mathbf{r}, t)$ the current density of the polarized vacuum: $\mathbf{j} = \frac{\partial \mathbf{P}}{\partial t} + \nabla \times \mathbf{M}$. The vacuum polarization $\mathbf{P}$ and magnetization $\mathbf{M}$ are derived from the Euler-Heisenberg Lagrangian [11]:

$$\begin{aligned} \mathbf{P}(\mathbf{r}, t) &= \frac{\eta}{4\pi} [2(E^2 - B^2)\mathbf{E} + 7(\mathbf{E} \cdot \mathbf{B})\mathbf{B}], \\ \mathbf{M}(\mathbf{r}, t) &= \frac{\eta}{4\pi} [-2(E^2 - B^2)\mathbf{B} + 7(\mathbf{E} \cdot \mathbf{B})\mathbf{E}], \end{aligned} \quad (2)$$

where $\eta = e^4/45\pi m^4$. $\mathbf{E}(\mathbf{r}, t)$ and $\mathbf{B}(\mathbf{r}, t)$ are the total electric and magnetic fields, respectively, consisting of the strong external field [marked by an index (0)] and the quantized probe field [marked by an index (p)]: $\mathbf{E}(\mathbf{r}, t) = \mathbf{E}^{(0)}(\mathbf{r}, t) + \mathbf{E}^{(p)}(\mathbf{r}, t)$, $\mathbf{B}(\mathbf{r}, t) = \mathbf{B}^{(0)}(\mathbf{r}, t) + \mathbf{B}^{(p)}(\mathbf{r}, t)$, $\mathbf{A}(\mathbf{r}, t) = \mathbf{A}^{(0)}(\mathbf{r}, t) + \mathbf{A}^{(p)}(\mathbf{r}, t)$. For a photon scattering (one incoming and one outgoing photon) only the term $\mathbf{j}^{(1)}\mathbf{A}^{(p)}(\mathbf{r}, t)$ of the interaction Hamiltonian contributes, with

$$\begin{aligned} \mathbf{j}^{(1)} &= \frac{\eta}{4\pi} \left\{ \frac{\partial}{\partial t} [-2\mathbf{B}^{(0)2}\mathbf{E}^{(p)} + 7(\mathbf{E}^{(p)}\mathbf{B}^{(0)})\mathbf{B}^{(0)}] \right. \\ &\quad \left. + 2\nabla \times [2(\mathbf{B}^{(0)}(\mathbf{B}^{(p)}\mathbf{B}^{(0)}) + \mathbf{B}^{(0)2}\mathbf{B}^{(p)})] \right\}, \end{aligned} \quad (3)$$

where $\mathbf{j}^{(l)}$ denotes the term of the current containing the probe field in the l th order. $\mathbf{j}^{(2)}\mathbf{A}^{(0)}(\mathbf{r}, t)$ can be responsible only for processes with a pair of photons in the initial or the final state.

First, we consider a photon scattering in a periodic field of a magnetic undulator of a linear polarization: $\mathbf{B}^{(0)}(\mathbf{r}) = \mathbf{B}_0 \cos \mathbf{q}\mathbf{r}$. In this case, the photon scattering probability per unit time reads:

$$\frac{dW}{d\Omega} = \rho_1 \omega^4 |M_{21} + M_{12}^*|^2 \mathcal{P}, \quad (4)$$

with the density of incoming photons ρ_1 , $\omega = k \equiv \omega_1 = \omega_2$, the phase-matching factor

$$\mathcal{P} = \left| \int_{(\mathcal{V})} e^{i\Delta \mathbf{k}\mathbf{r}} d^3\mathbf{r} \right|^2, \quad (5)$$

and

$$\begin{aligned} M_{21} &= \frac{\eta}{16\pi} \left\{ 4[(\mathbf{n}_1 \times \mathbf{e}_1)\mathbf{B}_0][(\mathbf{n}_1 \times \mathbf{e}_2)\mathbf{B}_0] \right. \\ &\quad + 7(\mathbf{e}_1\mathbf{B}_0)(\mathbf{e}_2\mathbf{B}_0) + \frac{8}{\omega} [(\mathbf{n}_1 \times \mathbf{e}_1)\mathbf{B}_0][\mathbf{e}_2(\mathbf{q} \times \mathbf{B}_0)] \\ &\quad \left. + \frac{4B_0^2}{\omega} (\mathbf{e}_2[\mathbf{q} \times (\mathbf{n}_1 \times \mathbf{e}_1)]) \right\}. \end{aligned} \quad (6)$$

The transition matrix element M_{21} simplifies when the incident photon momentum $\mathbf{k}_1$ is perpendicular to $\mathbf{B}_0$:

$$M_{21}^{\perp} = \frac{7}{8\pi} \eta B_0^2, \quad M_{21}^{\parallel} = \frac{\eta B_0^2}{2\pi} \cos \vartheta, \quad (7)$$

where $M_{21}^{\perp}$ corresponds to the transverse ($\mathbf{B}_1 \perp \mathbf{B}_0$) and $M_{21}^{\parallel}$ to the longitudinal polarization ($\mathbf{B}_1 \parallel \mathbf{B}_0$), and ϑ is the angle between $\mathbf{k}_1$ and $\mathbf{k}_2$. The space integration in Eq. (5) is carried out over the interaction volume $\mathcal{V}$, i.e., over the region of the overlap between the impinging laser beam with the undulator field. The factor $\mathcal{P}$ generates the phase-matching Bragg condition $\Delta \mathbf{k} = \mathbf{k}_2 - \mathbf{k}_1 - n\mathbf{q} = 0$ (with $n = 2$ in the case of a magnetic undulator) when the waves scattered from different spatial periods of the structure interfere constructively (Bragg interference). This can take place only at certain impinging angles of the probe wave:

$$2k \sin \frac{\vartheta}{2} = nq. \quad (8)$$

The scattered radiation is concentrated in the specular direction. Because of the Bragg interference, $\mathcal{P} \propto \mathcal{V}^2$ at exact phase-matching $\Delta \mathbf{k} = 0$ and the differential probability is proportional to $\mathcal{V}^2$. In particular, if the interaction region is rectangular with corresponding lengths L_x, L_y, L_z ,

$$\mathcal{P} = \left(\mathcal{V} \text{sinc} \frac{\Delta k_x L_x}{2} \text{sinc} \frac{\Delta k_y L_y}{2} \text{sinc} \frac{\Delta k_z L_z}{2} \right)^2. \quad (9)$$

where z axis is along the undulator wave-vector ($\mathbf{q}$) direction. The total probability integrated over angular distribution is

$$W = (2\pi)^2 \rho_1 \omega^2 |M_{21}|^2 \int L_{z'}^2(\mathbf{r}_{\perp}) d^2\mathbf{r}_{\perp}, \quad (10)$$

where $L_{z'}(\mathbf{r}_{\perp})$ is the length of the interaction region along the direction of $\mathbf{k}_2$ and $\mathbf{r}_{\perp}$ the coordinate transverse to $L_{z'}(\mathbf{r}_{\perp})$ [19]. Estimating $\int L_{z'}^2(\mathbf{r}_{\perp}) d^2\mathbf{r}_{\perp} \approx \bar{L}_{x'} \bar{L}_{y'} \bar{L}_{z'}$, with $\bar{L}_{x'}, \bar{L}_{y'}, \bar{L}_{z'}$ being the mean lengths of the interaction region along and transverse to $\mathbf{k}_2$, the number of scattered photons per unit time and volume is derived:

$$\frac{d\mathcal{N}}{dt dV} \approx \mathcal{G} \mathcal{W}(\vartheta) \alpha m^4 \frac{B_0^4}{B_{cr}^4} \frac{E_0^2}{E_{cr}^2} \varepsilon, \quad (11)$$

where E_0 is the amplitude of the probe field, $\mathcal{W}(\vartheta) \equiv |M_{21}|^2/(\eta B_0^2)^2$, $\mathcal{G} = (1/45)^2$, $\varepsilon \equiv \bar{L}_{z'}/\lambda$ is the enhancement factor due to Bragg interference. The comparison of the Bragg scattering with the stimulated light-by-light scattering [13,15], shows that the Bragg scattering can be larger by a factor $\bar{L}_{z'}/w_0$, with the laser beam waist size w_0 , which can amount to an order of magnitude. Note that Eq. (11) is valid when the probe beam is rather monochromatic and has a low angular spread. The bandwidth ($\Delta\omega_1$) and the angular spread ($\Delta\vartheta$) of the probe beam should be limited to fulfill the Bragg condition Eq. (8) within the energy uncertainty $\sim 1/\tau$, with the interaction time τ : $\Delta\omega_1 \leq (2\pi/\tau)D_{\omega_1} = (2\pi/\tau)(2/n^2)\sin^2(\vartheta/2)$ and $\Delta\vartheta \leq (2\pi/\tau)D_{\vartheta} = (2\pi/\tau)(1/q \cos(\vartheta/2))$, where $D_{\omega_1} = |\partial(\omega_1 - \omega_2)/\partial\omega_1|$ and $D_{\vartheta} = |\partial(\omega_1 - \omega_2)/\partial\vartheta|$. The latter impose restrictions on the enhancement factor ε . The largest interaction length $\bar{L}_{z'}$, equal to the undulator length L_u , is possible at $\vartheta \sim \pi$ when $k \sim q$. Then, the enhancement factor, determined by the number N_u of undulator periods $\varepsilon \sim N_u$, is restricted by the probe bandwidth $\varepsilon \leq \omega/\Delta\omega_1$.

Now let us consider Bragg scattering when the periodic structure is formed using a set of N focused laser beams propagating parallel to each other:

$$\mathbf{E}^{(0)} = \mathbf{E}_0 \sum_{n=1}^N f(x, y, z - nd) \times \cos[\omega_L t - k_L x + \varphi(x, y, z - nd)], \quad (12)$$

where $f(x, y, z) = (\sqrt{w_y w_z}/\sqrt{w_y(x)w_z(x)})e^{-y^2/w_y^2(x) - z^2/w_z^2(x)}$, $\varphi(x, y, z) = -\frac{k_L y^2}{2R_y(x)} - \frac{k_L z^2}{2R_z(x)} + \frac{1}{2}t a n^{-1} \frac{x}{x_{0y}} + \frac{1}{2}t a n^{-1} \frac{x}{x_{0z}}$, $w_y(x) = w_y \sqrt{1 + \frac{x^2}{x_{0y}^2}}$, $w_z(x) = w_z \sqrt{1 + \frac{x^2}{x_{0z}^2}}$, $R_y(x) = x + \frac{x_{0y}^2}{x}$, $R_z(x) = x + \frac{x_{0z}^2}{x}$, $x_{0y} = \frac{k_L w_y^2}{2}$ and $x_{0z} = \frac{k_L w_z^2}{2}$. The laser beams are elliptic Gaussian [24] with $w_y > w_z$ to allow for a large overlap with the probe beam (due to a large w_y) and simultaneously to fulfill the Bragg condition (due to a small w_z). The distance between the adjacent beams d is assumed to be larger than the waist size of a single beam w_z ; therefore, the superposition of fields of different beams is negligible. A slow variation of the beam profile $f(\mathbf{r})$ is also assumed: $|\nabla f(\mathbf{r}) \times \mathbf{E}_0| \ll k_L f(\mathbf{r}) E_0$ and $|\nabla f(\mathbf{r})| \ll k_L f(\mathbf{r})$. In this case $\mathbf{B}_0 = \mathbf{n}_L \times \mathbf{E}_0$, where $\mathbf{n}_L = \mathbf{k}_L/k$ and $\mathbf{k}_L$ is the laser wave vector. The current calculated as

$$\mathbf{j}^{(1)} = \frac{\eta}{4\pi} \left(F^2(\mathbf{r}, t) \frac{\partial \mathbf{R}_1}{\partial t} + F^2(\mathbf{r}, t) \nabla \times \mathbf{R}_2 + \nabla F^2(\mathbf{r}, t) \times \mathbf{R}_2 \right), \quad (13)$$

where $\mathbf{R}_1 \equiv 4[(\mathbf{E}^{(p)} \mathbf{E}_0) - (\mathbf{B}^{(p)} \mathbf{B}_0)]\mathbf{E}_0 + 7[(\mathbf{E}^{(p)} \mathbf{B}_0) + (\mathbf{B}^{(p)} \mathbf{E}_0)]\mathbf{B}_0$, $\mathbf{R}_2 \equiv \mathbf{R}_1 \times \mathbf{n}_L$ and $F^2(\mathbf{r}, t) \equiv \sum_{l,n}^N$

$f(x, y, z - ld)f(x, y, z - md) \cos[\omega_L t - k_L x + \varphi(x, y, z - ld)] \cos[\omega_L t - k_L x + \varphi(x, y, z - md)]$. We consider only elastic scattering of photons. However, in this situation inelastic scattering can also take place with absorption or emission of additional laser photons: $\omega_1 = \omega_2 \pm 2\omega_L$, which can be enhanced by the Bragg interference when coherence is met: $\mathbf{k}_1 = \mathbf{k}_2 \pm 2\mathbf{k}_L + \mathbf{q}$. In the case of elastic scattering, we average the current over the fast oscillations of nonresonant terms which does not fulfill the Bragg condition (i.e., replacing $\cos^2(\omega_L t - k_L x + \varphi) \rightarrow 1/2$). The photon scattering probability in the setup of focused laser beams is given by Eq. (4) with the following matrix element:

$$M_{21} = \frac{\eta}{8\pi} \left\{ 4[(\mathbf{e}_1 \mathbf{E}_0) + \{(\mathbf{e}_1 \times \mathbf{n}_1)\mathbf{B}_0\}] \times \left[(\mathbf{e}_2 \mathbf{E}_0) + \left\{ \mathbf{e}_2 \left[\left(\mathbf{n}_1 - n \frac{\mathbf{q}}{k} \right) \times \mathbf{B}_0 \right] \right\} \right] + 7[(\mathbf{e}_1 \mathbf{B}_0) - ((\mathbf{e}_1 \times \mathbf{n}_1)\mathbf{E}_0)] \times \left[(\mathbf{e}_2 \mathbf{B}_0) - \left(\mathbf{e}_2 \left[\left(\mathbf{n}_1 - n \frac{\mathbf{q}}{k} \right) \times \mathbf{E}_0 \right] \right) \right] \right\}. \quad (14)$$

The phase-matching factor $\mathcal{P}$, after neglecting corrections proportional to the small diffraction parameter, is:

$$\mathcal{P} = \frac{(\pi L_x w_y w_z)^2}{16} e^{-(\delta k_y^2 w_y^2)/4 - (\delta k_z^2 w_z^2)/4} \times \text{sinc}^2 \frac{\delta k_x L_x}{2} \frac{\sin^2 \frac{\delta k_z N d}{2}}{\sin^2 \frac{\delta k_z d}{2}}, \quad (15)$$

where $\delta \mathbf{k} = \mathbf{k}_2 - \mathbf{k}_1$ and L_x is the interaction length along the axis of the laser beams consisting the Bragg structure. The last multiplicative term in Eq. (15) has maxima at the Bragg condition $\delta k_z = nq$ with a bandwidth $\Delta(\delta k_z) \sim 2q/N$, where $q = 2\pi/d$ and n is an integer number. The exponential damping factor determines the maximal value for $\delta k_z \leq 2/w_z$, while the Bragg condition determines the harmonic number $n \leq d/\pi w_z$ (the damping factor is $e^{-n^2 q^2 w_z^2/4}$). The strong laser field is linearly polarized. Two cases are possible: either $\mathbf{B}_0$ (B field) or $\mathbf{E}_0$ (E field) is perpendicular to the scattering plane formed by the wave vectors $\mathbf{k}_1$ and $\mathbf{k}_2$. The scattering matrix element for different polarizations of the probe wave reads:

$$M_{21}^{B\parallel} = \frac{4\eta E_0^2}{\pi} \sin^4 \frac{\vartheta}{4} \left(1 - 4\cos^2 \frac{\vartheta}{4} \right),$$

$$M_{21}^{B\perp} = \frac{7\eta E_0^2}{\pi} \sin^4 \frac{\vartheta}{4}, \quad M_{21}^{E\parallel} = \frac{4\eta E_0^2}{\pi} \sin^4 \frac{\vartheta}{4}, \quad (16)$$

$$M_{21}^{E\perp} = \frac{7\eta E_0^2}{\pi} \sin^4 \frac{\vartheta}{4} \left(1 - 4\cos^2 \frac{\vartheta}{4} \right),$$

where the longitudinal (transverse) polarization means $B_1 \parallel (\perp) B_0$ in the case of the B field and $E_1 \parallel (\perp) E_0$ for the E field. The total number of scattered photons per unit volume and time in the laser beam setup is given by the same Eq. (11) with $\mathcal{G} = e^{-n^2 q^2 w_z^2/4} (w_z q)^2 / (\sqrt{2\pi}(360)^2)$ and replacing $B_{cr} \rightarrow E_{cr}$, $B_0 \rightarrow E_0$. The angular

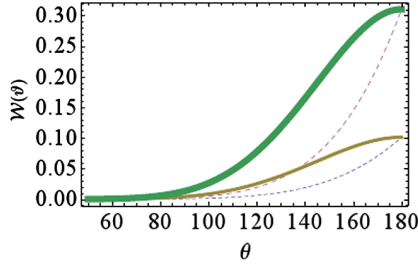


FIG. 2 (color online). The dependence of the scattering probability on the probe impinging angle [the factor $\mathcal{W}(\theta)$]: (solid line) B -field case, the upper curve—transverse polarization, the lower curve—longitudinal polarization; (dashed line) E -field case, the upper curve—longitudinal polarization, the lower curve—transverse polarization.

dependence factor $\mathcal{W}(\theta)$ is shown in Fig. 2. The enhancement factor in this case is

$$\varepsilon = Nd/\lambda, \quad \text{if } kw_y^2/4 \gtrsim Nd$$

$$\varepsilon = 2\pi\sqrt{Nd/\lambda}(w_y/\lambda), \quad \text{otherwise,} \quad (17)$$

with the wavelength λ . When $kw_y^2/4 \lesssim Nd$, the emission angular width becomes restricted by the width of the energy uncertainty, as the Bragg condition should be fulfilled simultaneously. We estimate the yield of scattered photons in the setup of multiple laser beams. Employing a laser field with an intensity of $I = 2.3 \times 10^{22}$ W/cm², $\lambda = 1$ μ m, pulse duration of $\tau = 100$ fs, $w_y \approx 3\lambda$, $w_z \sim \lambda$, $d \approx \pi w_z$ and $N = 10$, the Bragg scattering angle is close to π at $n = 2$ resonance, and the number of scattered photons per pulse is

$$\mathcal{N} = \frac{\alpha m^4 (2\pi)^{3/2}}{(360)^2 e^4} \left(\frac{I}{I_{cr}} \right)^3 (w_y L_x N d \tau) \frac{Nd}{\lambda} \mathcal{W}(\vartheta) \approx 4.8. \quad (18)$$

At a fixed intensity of the waves, the Bragg scattering setup provides an enhancement of the number of scattered photons with respect to the stimulated photon-photon scattering by a factor proportional to N^2 , see Eq. (18). However, usually the available power is restricted in an experiment. At a fixed total intensity I_{tot} , $I = I_{tot}/(N + 1)$ in the Bragg scattering setup, while $I = I_{tot}/3$ in the case of stimulated photon-photon scattering with three laser beams [15]. Accordingly, the enhancement factor of the Bragg setup over the stimulated photon-photon scattering will be $3^3 N^2/(N + 1)^3$ which will provide approximately 2 times enhancement at $N = 10$. Note that N should be large enough to confine the reflected wave in a small divergence angle along the specular direction, $\delta\theta \sim |dk_z - nq|/k \sim \lambda/Nd \ll 2/kw_z$, and to distinguish the scattered photons from the beams of the Bragg structure. At the same time N is restricted to maintain the necessary large value of I/I_{cr} .

Concluding, we have shown for the first time how the coherence effects arising in spatially structured vacuum in

strong periodic fields can enhance the vacuum polarization effects. Similar enhancement effects will exist in all types of inelastic light-by-light scattering and other processes based on spatially modulated vacuum polarization.

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